

noise suppression filters in satellites and defence systems. Applications for the inductors include low-profile, high-current power supplies, point-of-load converters, flight-ready solar inverters, high-current filters and DC/DC converters in distributed power systems.

Offered in five case sizes, the inductors feature high efficiency with typical DCR from 0.56mΩ to 240mΩ and a wide range of inductance values from 0.22μH to 100μH. Based on IHLP technology, the devices provide rated currents up to 100A and handle high transient current spikes without saturation. The inductors offer excellent inductance and saturation stability over the complete operating temperature range.

Vishay Intertechnology Inc.

www.vishay.com

Digital-to-analogue converter

Analog Devices has introduced AD5758 digital-to-analogue converter (DAC). It incorporates the company's second-generation dynamic power control (DPC) technology to enable high-density AOUT modules without requiring de-rating—the need to turn off channels due to thermal build up. This single-channel current/voltage DAC with DPC is designed for channel-to-channel isolated industrial applications in factory automation, process automation and motor control. It is the company's lowest-power industrial DAC.

AD5758's 5mm x 5mm size creates the smallest footprint in the industry. Its robust construction requires fewer external protection components, supporting smaller, lower-cost designs. It is engineered with advanced diagnostics that provide visibility to system performance for higher reliability and planned maintenance schedules. AD5758 also includes output fault protection circuits to safeguard the DAC in the event of a mis-wire incident.

Analog Devices Inc.

www.analog.com

TEST & MEASUREMENT

Thermal imaging system

FLIR C3 is a full-featured, pocket-size thermal camera designed to be your

go-to tool for electrical/mechanical and building applications.

It is great for finding hidden hotspots, structural defects, plumbing issues and more. C3's key features include MSX real-time image enhancement, area maximum or minimum temperature measurement and Wi-Fi connectivity, so you can quickly get to the job of finding hidden problems, sharing images and documenting repairs.

Other features include:

- Light, slim profile that fits comfortably in any work pocket
- Brilliant 7.6cm (3-inch) intuitive touchscreen with auto orientation for easy viewing
- Radiometric image storing 4800 pixels, capable of capturing thermal measurements from -10°C to 150°C
- Accuracy of $\pm 2^\circ\text{C}$
- Focus free
- Thermal sensitivity $< 0.10^\circ\text{C}$

FLIR Systems India Pvt Ltd

www.flir.in

AM/FM modulation option

All Zurich Instruments lock-in amplifiers can now be equipped with amplitude and frequency modulation capabilities with the new AM/FM modulation option. This option enables coherent generation and detection of signals with multiple frequency components, such as amplitude- and frequency-modulated signals. Up to three frequencies can be coherently combined to generate fully-configurable AM signals.

Zurich Instruments AG

www.zhinst.com

INTERNET OF THINGS

MEMS sensors

STMicroelectronics is extending its commitment to drive advanced automation and the industrial Internet of Things (IIoT) by adding high-stability MEMS sensors with 10-year product-longevity

assurance. The sensors, to be made available during 2018 begin with IIS3D-HHC, a 3-axis accelerometer optimised for high measurement resolution and stability to ensure accuracy over time and temperature.

IIS3DHHC targets precision inclinometers in antenna-positioning mechanisms for communication systems, structural health monitoring equipment for keeping buildings and bridges safe, and stabilisers or levellers for a wide variety of industrial platforms. Its long-term accuracy and robustness are also ideal for high-sensitivity tilt and security sensors, as well as image stabilisation in high-end digital still cameras.

STMicroelectronics India

www.st.com

Connectivity boards

Arduino has launched two wireless connectivity boards to help streamline IoT development.

The first is MKR Wi-Fi 1010, which offers low power consumption. It has been designed not only to speed up and simplify the prototyping of Wi-Fi-based IoT applications, but also to be embedded in production IoT applications that require Wi-Fi connectivity. The board is an evolution of the existing Arduino MKR1000, but now comes equipped with an ESP32-based module manufactured by u-blox. A significant feature is MKR1010's two standalone programmable processors: first based on ARM processor core technology and second based on a dual-core Espressif IC. This makes it a high-performance solution that can distribute the workload across its dual-processor system.

The second board is MKR NB 1500, which employs the new low-power NB-IoT (Narrow-Band IoT) standard, designed to work over mobile/LTE networks. When combined with the ease of use of Arduino ecosystem, MKR1500 becomes an excellent choice for applications in remote areas, such as on-field monitoring systems and remote-controlled LTE-enabled modules.

Arduino

www.arduino.cc

